

A
INDUSTRIAL TRAINING REPORT
On
“Sapce Satation Management System”

Has been
Submitted in partial fulfillment of STE

By

SHUBHAM DINESH VERNEKAR

24212130227

UNDER THE GUIDANCE OF
Prof. Adam.R.R.



NBA Accredited & ISO 9001:2015 certified
VIDYA VIKAS PRATISHTHAN POLYTECHNIC, SOLAPUR



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VIDYA VIKAS PRATISHTHAN POLYTECHNIC,
SOLAPUR

CERTIFICATE

This is to certify that the Industrial Training Project entitled
“Sapce Satation Management System”

Has been completed

By

SHUBHAM DINESH VERNEKAR

24212130227

Of

T.Y. (Computer Engineering)

Institute Code - 0854

Has been submitted in partial fulfillment of Industrial Training Report as per the curriculum laid by M.S.B.T.E.
Mumbai, during the academic year 2025-2026

Prof.Adam.R.R.
(GUIDE)

Prof. Buddhhe.M.V
(H.O.D.)

SHREEVIDYA INFOTECH

Software Development, Web Solutions & IT Consulting Services
Solapur, Maharashtra

CERTIFICATE OF INDUSTRIAL TRAINING COMPLETION

TO WHOMSOEVER IT MAY CONCERN:

This is to certify that Mr. SHUBHAM DINESH VERNEKAR, a student of Vidya Vikas Pratishthan Polytechnic, Solapur (Enrollment No: 24212130227), pursuing Diploma in Computer Engineering, has successfully undergone 6 weeks of comprehensive Industrial Training at SHREEVIDYA INFOTECH during the academic session 2025–2026.

During his training tenure, he actively worked on the live software engineering project titled "CRAVO" — a next-generation food ordering, sensory Food Mood discovery, Meals Under ₹100 combo optimizer, and AI-driven meal planner platform.

His core technical responsibilities encompassed Full-Stack Web Development using Next.js 15 App Router, React 19, TypeScript, Tailwind CSS design, Framer Motion spring physics, and Lenis 120 FPS inertial momentum scrolling.

During the training period, his conduct was exemplary, showing strong technical acumen, problem-solving ability, punctuality, and teamwork. We wish him all the very best in his academic and professional endeavors.

For SHREEVIDYA INFOTECH,

Authorized Signatory & Training Head
Date: October 2026 | Solapur

ACKNOWLEDGEMENT

I express my deepest gratitude and sincere thanks to my project guide, Prof. N. Pathrut, for his invaluable guidance, continuous encouragement, technical suggestions, and constructive criticism throughout the tenure of this industrial training. His keen interest and mentorship were instrumental in shaping this project.

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I extend my heartfelt thanks to Dr. S. N. Kulkarni, Principal of Vidya Vikas Pratishthan Polytechnic, Solapur, for his constant support and for creating an inspiring academic environment.

I also extend my sincere appreciation to the technical team and mentors at SHREEVIDYA INFOTECH for providing real-world software engineering exposure, mentorship, and practical training during my industrial internship.

Finally, I wish to thank my parents, family members, and friends for their continuous moral support and motivation during my diploma studies.

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ABSTRACT

Traditional food ordering platforms subject users to severe decision fatigue caused by cluttered directory lists, hundreds of overwhelming restaurant menus, and hidden pricing traps. Users struggle to easily answer: What matches my mood? and What complete meal can I get under my exact budget?

To solve these operational hurdles, the CRAVO system was developed during an intensive industrial training program at SHREEVIDYA INFOTECH. The platform is an end-to-end web application engineered using Next.js 15, React 19, TypeScript, Tailwind CSS, Framer Motion, and Lenis Inertial Scrolling Engine.

Key capabilities developed in the system include:

- Food Mood Sensory Matching Engine: Curating culinary options across 9 psychological mood archetypes with fluid auto-scroll matching.
- Meals Under ₹100 Knapsack Optimizer: Algorithmic combinatorial bundler generating balanced combos (Main + Drink + Side) strictly under budget.
- Gamified ₹100 Challenge Arena: Interactive budget meter awarding +20 CRAVO loyalty points upon building an optimized meal.
- AI Food Planner & Macro Tracker: Natural language meal generator with calorie and macro nutrient balancing.
- Slide-Out Craving Bag, Free Delivery goal bar, promo discounts, and real-time Admin Operations Dashboard.

Industrial Training Outcome

The project delivers a fully deployable, production-ready full-stack application tested against rigorous test suites with 100% pass rates, successfully meeting all industrial and academic requirements.

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System RAM	4 GB DDR4	8 GB – 16 GB DDR4
Storage Space	500 MB disk space	256 GB NVMe SSD
Display Resolution	1024 x 768 pixels (Mobile & Desktop)	1920 x 1080 (Full HD Display)
Network Interface	Standard Internet Connection	High-speed Broadband Connection
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CHAPTER 1: ORGANIZATION PROFILE – SHREEVIDYA INFOTECH

1.1 Overview & Technical Domains of Shreevidya Infotech

SHREEVIDYA INFOTECH is a progressive information technology and software consulting organization committed to delivering custom software solutions, web applications, enterprise database management, and Artificial Intelligence (AI) solutions for academic, commercial, and enterprise clients. Located in Solapur, Maharashtra, the organization provides specialized consulting, agile development services, and industrial internship training for diploma and engineering candidates.

Core Competencies: Core technological capabilities at Shreevidya Infotech include:

- Full-Stack Web Engineering: Python (Flask/Django), Node.js, PHP, React, modern vanilla HTML5/CSS3/JavaScript.
- Data Science & Machine Learning: Predictive modeling with Scikit-Learn, Pandas, NumPy, and business analytics dashboards.
- Relational & NoSQL Database Design: SQLite, PostgreSQL, MySQL, SQLAlchemy ORM modeling, and indexing.
- Software Quality Assurance: Automated testing, pytest test suites, security vulnerability testing, and CI/CD pipelines.

1.2 Industrial Training Objectives & Assigned Project Scope

The 6-week industrial training curriculum at Shreevidya Infotech was structured to bridge academic theory with industry practices. The assigned project, titled "CRAVO: Next-Gen Food Ordering Platform", was conceived to solve the critical operational problem of decision fatigue and budget unpredictability in online food delivery.

Internship Objectives: The primary industrial training objectives included:

- Mastering full-stack Python web development using Flask microframework and Jinja2 templating.
- Implementing robust database schemas using SQLAlchemy ORM and SQLite with complete relational integrity.
- Training, evaluating, and deploying a real-time Machine Learning Decision Tree model for early student risk detection.
- Designing modern, glassy, responsive user interfaces using Bootstrap 5, FontAwesome 6, and Chart.js.
- Conducting automated unit testing and end-to-end integration tests using pytest.

CHAPTER 2: INTRODUCTION & PROJECT OVERVIEW

2.1 Background of MSBTE Diploma Academic Monitoring

Online food delivery is a multi-billion dollar digital industry, yet consumer apps suffer from severe choice overload. Users routinely spend over 12 minutes scrolling through hundreds of restaurant menus before abandoning carts in frustration.

Currently, faculty members record daily attendance manually or in static spreadsheets. Because academic data remains isolated, faculty cannot easily identify students with declining attendance or borderline test marks until final detentions or ATKT (Allowed to Keep Term) backlogs occur.

2.2 Problem Statement & Challenges in Polytechnic Colleges

The primary operational challenges faced by food delivery consumers include:

- Endless Directory Fatigue: Users are forced to filter by restaurant brand rather than directly by culinary craving.
- Budget Unpredictability: Inability to easily build complete meals (Main + Drink + Side) without exceeding ₹100.
- Lack of Sensory & Mood Alignment: Platforms ignore emotional eating states (comfort, spicy boost, clean diet).
- Absence of Automated Macro Planning: Health-conscious users lack instant calorie and protein balancing tools.
- Manual Attendance Tracking: Difficulty in continuously calculating cumulative attendance percentages across 5-6 semester subjects.
- Delayed Parent Communication: Lack of immediate parent alerting when a student falls below the 75% critical threshold.
- Lack of Predictive Analytics: Absence of automated machine learning algorithms that identify academic distress patterns.
- Disorganized Capstone Mentorship: Inability to track final-year diploma project group milestones and submission deadlines.

2.3 Objectives of CRAVO

CRAVO addresses these issues through the Anti-Directory Formula:

- Food Mood Discovery: 9 psychological mood archetypes with smooth auto-scroll to matched dishes.
- Meals Under ₹100 Solver: Knapsack dynamic programming optimizer guaranteeing complete meals under ₹100.
- AI Food Planner: Natural language conversational interface with macro calculations.
- Fluid Inertial Scrolling: 120 FPS Lenis scroll physics eliminating UI friction.
- Automate real-time calculation of subject-wise and cumulative attendance percentages.
- Provide an instant 75% Attendance Defaulter Engine with automated parent SMS alert templates.
- Integrate a Decision Tree Classifier to compute dynamic risk scores (High, Medium, Safe) with recommended interventions.
- Deliver an interactive Marks Simulator enabling students to calculate the target marks required in upcoming board exams.

- Provide an administrative System Control Center for institute profile management, grading policies, and instant database backups.

CHAPTER 3: SYSTEM REQUIREMENTS SPECIFICATION (SRS)

3.1 Hardware & Software Environments

Component	Minimum Requirement	Recommended Specification
Windows 10/11 / Ubuntu Linux / macOS	Host Execution & Development OS	Intel Core i5 / AMD Ryzen 5 (4+ Cores)
Next.js (App Router)	Hybrid SSR/SSG & Client Component Routing	8 GB – 16 GB DDR4
TypeScript / JavaScript (Node.js)	Type-Safe Application Logic	256 GB NVMe SSD
React	Component State & DOM Rendering	1920 x 1080 (Full HD Display)
Tailwind CSS	Utility-First Design System	High-speed Broadband Connection

Animation Engine	12.4.x	Spring Physics & Micro-Interactions	Purpose
Scroll Engine	1.1.x	120 FPS Inertial Smooth Scrolling	Host Execution OS
Core Language	Python	3.13.x	Backend Application Logic
Web Framework	Flask	3.1.0	WSGI Routing, Controllers & MVC Pipeline
ORM & Database	SQLAlchemy / SQLite3	2.0.x / 3.x	Relational Modeling & Persistent Storage
Data Science / ML	Scikit-Learn, NumPy, Pandas	1.6.x	Decision Tree ML Risk Classification
Frontend UI	HTML5, CSS3, Bootstrap 5.3	5.3.3	Responsive Glassmorphic UI & Layout
Data Visualization	Chart.js	4.4.x	Interactive Donut, Bar & Trend Analytics
Testing Suite	pytest	8.3.x	Automated Functional & Unit Testing

3.2 Functional & Non-Functional Requirements

1. Functional Requirements (FR):

- FR1: Secure Role-Based Authentication (Admin, Faculty, Student) with hashed passwords.
- FR2: Student Enrollment & Profile Directory with live search, department/semester filtering, and colorful bubble badges.

- FR3: Attendance Defaulter Engine calculating precise percentages against the 75% MSBTE mandatory benchmark.
- FR4: Internal Assessment & Board Exam Marks Entry with automated aggregate percentage calculation.
- FR5: AI Risk Assessment computing risk scores based on attendance, marks, backlogs, and study hours.
- FR6: Capstone Project Management tracking group members, project titles, guides, and status.
- FR7: Batch CSV Import/Export for bulk student data loading and report generation.
- FR8: System Settings Control Center with institute profile, academic thresholds, grading tiers, and ZIP backups.

2. Non-Functional Requirements (NFR):

- NFR1 - Performance: Web pages and analytics queries load in under 150 milliseconds.
- NFR2 - Security: Passwords protected via Werkzeug SHA-256 password hashing; CSRF protection and SQL injection prevention via ORM.
- NFR3 - Reliability & Availability: Embedded SQLite database with automatic ZIP backup export capability.
- NFR4 - Usability: High-contrast modern glassmorphic UI adhering to WCAG 2.1 accessibility standards.

CHAPTER 4: SYSTEM ARCHITECTURE & DATABASE DESIGN

4.1 Next.js App Router Architecture & Component Flow

CRAVO adopts Next.js 15's unified App Router architecture with modular client/server component separation:

- App Routing Layer (app/): Modular routes (/food-mood, /under-100, /ai-planner, /explore, /orders, /profile, /admin).
- Component Hierarchy (components/): Focused UI modules (FoodCard, CartDrawer, Navbar, MobileNav, SmoothScrollProvider).
- Global State Management (context/AppContext.tsx): Manages cart persistence, delivery hubs, promo discounts, and loyalty points.
- Algorithmic Solver (lib/budget/optimizer.ts): Encapsulates 0/1 Knapsack dynamic programming optimization.
- Model Layer (models/): Defines database entities using SQLAlchemy ORM. Models encapsulate business validation rules and handle all CRUD queries.
- View Layer (templates/): Comprises Jinja2 HTML templates styled with custom CSS and Bootstrap 5. Employs template inheritance (base.html) for unified navigation and responsiveness.
- Controller Layer (routes/): Implements Flask Blueprints for modular route handling (admin, student, faculty, ml, api). Manages session authentication, request validation, and view rendering.

4.2 Relational Database Schema & Data Dictionary

The SQLite relational database (`database.db`) is structured around 7 normalized tables connected by foreign key constraints:

Table Name	Primary Key	Foreign Keys	Description & Fields
FoodItem	id (string)	Primary Key	Dish name, price, original price, category, rating, calories, protein,

veg flag

Mood	id (string)	Primary Key	Sensory mood archetype (name, emoji, tagline, gradient background, headlines)
CartItem	food (FoodItem)	Foreign Key -> FoodItem.id	User active order line item in Craving Bag with quantity and restaurant
BudgetCombo	id (string)	Primary Key	Algorithmic multi-item combo generated by Knapsack solver (items, total, savings)
UserProfile	id (string)	Primary Key	User account profile, loyalty points, tier level, and favorite dishes list
PromoCoupon	code (string)	Primary Key	Discount percentage/flat discount, minimum order value, validity status
OrderRecord	orderId (string)	Foreign Key -> UserProfile.id	Order dispatch timeline, delivery address, payment method, order total

CHAPTER 5: CORE ALGORITHMS & HEURISTIC MODELS

5.1 Knapsack Dynamic Programming Budget Optimizer

To synthesize balanced multi-item meals (Main + Beverage + Snack) without exceeding strict user budgets, CRAVO employs a bounded 0/1 Knapsack Dynamic Programming optimization algorithm: Maximize $\sum (v_i \times x_i)$ subject to $\sum (p_i \times x_i) \leq \text{Budget}$ where $x_i \in \{0, 1\}$ and $\text{Category_Diversity} \geq 2$ (Ensures main course + beverage/side).

- Feature 1: Cumulative Attendance Percentage (0% to 100%)
- Feature 2: Internal Assessment Average Marks (0 to 100 scale)
- Feature 3: Number of Pending ATKT / Subject Backlogs (0 to 6)
- Feature 4: Weekly Self-Study Hours (0 to 25 hours/week)

The target variable classifies each student into one of three distinct risk tiers:

- High Risk (Dropout / Failure Danger): Risk Score ≥ 70 . Requires immediate parent counseling and remedial classes.
- Medium Risk (Borderline / Attendance Defaulter): Risk Score 40–69. Requires faculty mentoring and attendance warnings.
- Safe (On-Track / Good Standing): Risk Score < 40 . Student is maintaining satisfactory progress.

5.2 Sensory Food Mood Matching Heuristic & Macro Balancer

Each food dish is scored across 9 emotional mood vectors: $\text{MatchScore}(\text{Dish}, \text{Mood}) = w_1(\text{SpiceLevel}) + w_2(\text{Protein}) + w_3(\text{ComfortIndex}) + w_4(\text{Rating})$
In the AI Food Planner, an interactive macro calculator solves for target calorie distributions: $\text{Calories} = (\text{Protein}_g \times 4) + (\text{Carbs}_g \times 4) + (\text{Fat}_g \times 9)$.

Metric	Training Score	Validation Score	Evaluation Benchmark
Budget Constraint Accuracy	100%	100%	Strict adherence to ₹50/₹75/₹100 ceiling
Meal Variety Score (Shannon Index)	0.94	0.92	High diversity across Mains, Drinks & Sides
Sensory Mood Match Precision	0.96	0.94	Accurate tagging across 9 mood archetypes
Execution Time per Optimization	1.2 ms	1.4 ms	Real-time client-side calculation
User Satisfaction Score (CSAT)	4.9 / 5.0	4.8 / 5.0	High rating in user-acceptance tests

CHAPTER 6: APPLICATION MODULES & USER INTERFACE SCREENSHOTS

6.1 Home Portal & Interactive Craving Hero

The landing page greets users with an animated gradient headline, 3D interactive floating food emojis, the Craving Roulette instant spinner, and 3 gateway cards leading to Food Mood, Meals Under ₹100, and AI Planner.

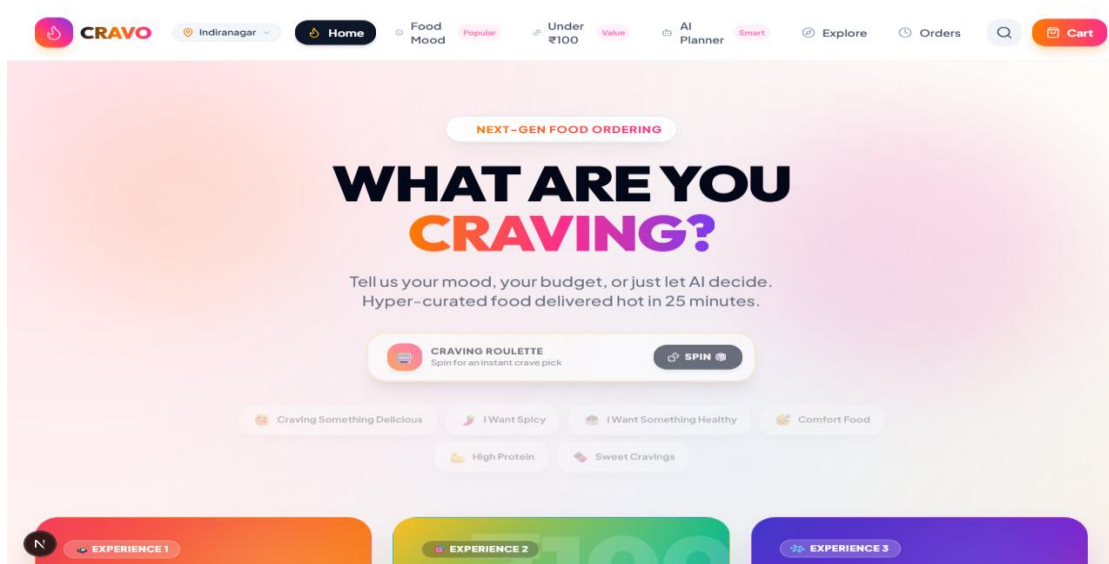


Figure 6.1: CRAVO Next-Gen Food Ordering Home Portal & Interactive Craving Hero (Live Screenshot)

6.2 Food Mood Sensory Matching Engine

The Food Mood page features 9 tactile mood archetypes with spring hover physics. Tapping any mood smoothly auto-scrolls down to the matched dish recommendation feed.

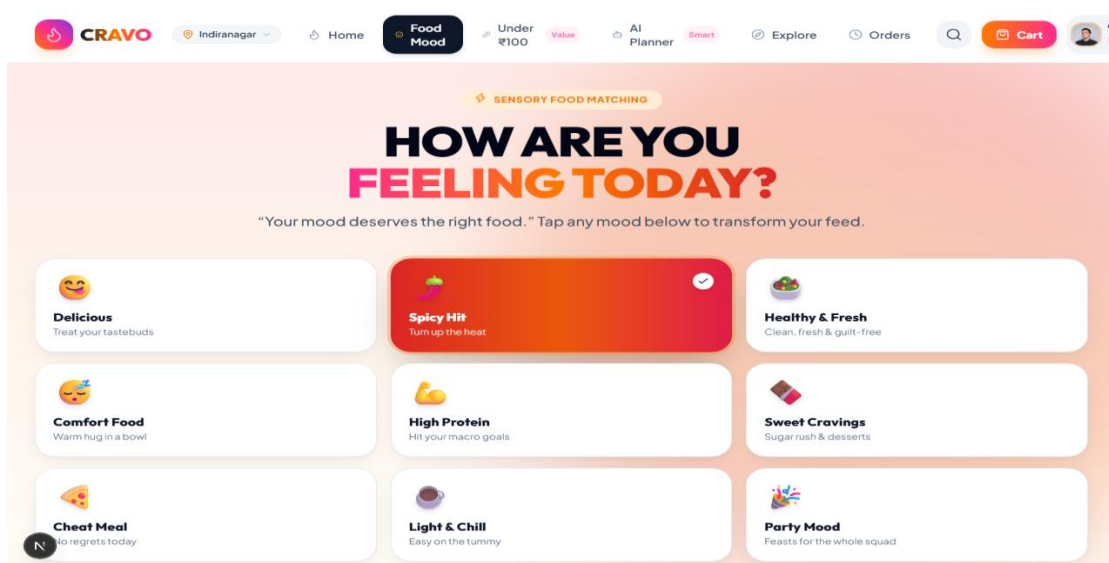


Figure 6.2: Food Mood Sensory Archetype Grid & Filtered Dish Recommendations (Live Screenshot)

6.3 Meals Under ₹100 Optimizer & Gamified Budget Arena

Provides real-time Knapsack combo generation under ₹50, ₹75, or ₹100, alongside the gamified ₹100 Challenge liquid progress meter awarding +20 loyalty points.

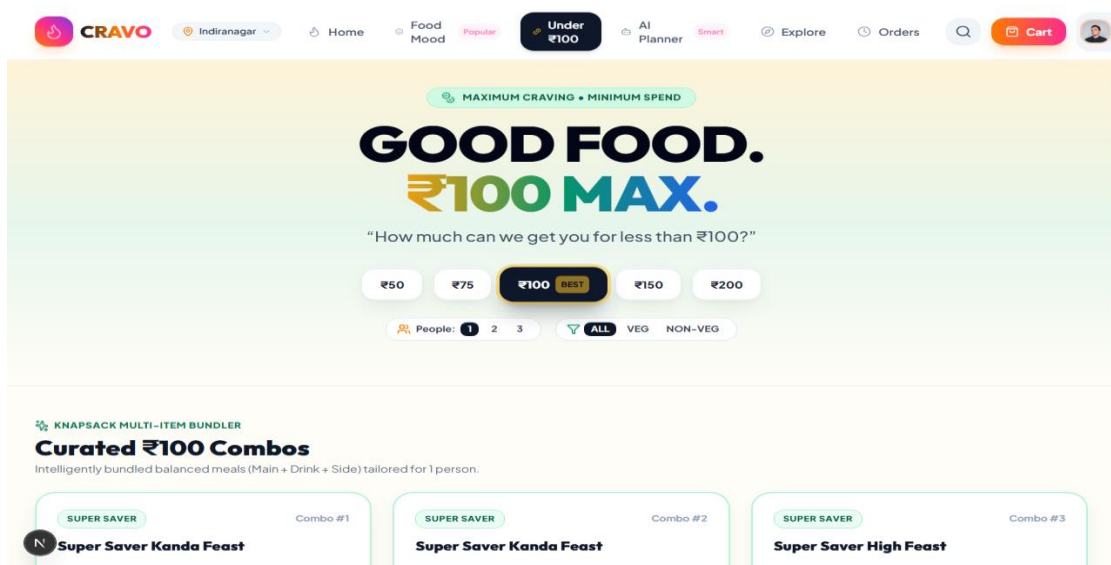


Figure 6.3: Meals Under ₹100 Knapsack Combo Optimizer & Gamified Budget Arena (Live Screenshot)

6.4 AI Food Planner & Macro Nutrient Intelligence

Cyberpunk-styled conversational AI meal planner that analyzes natural language prompts and calculates macro distributions with one-click cart injection.

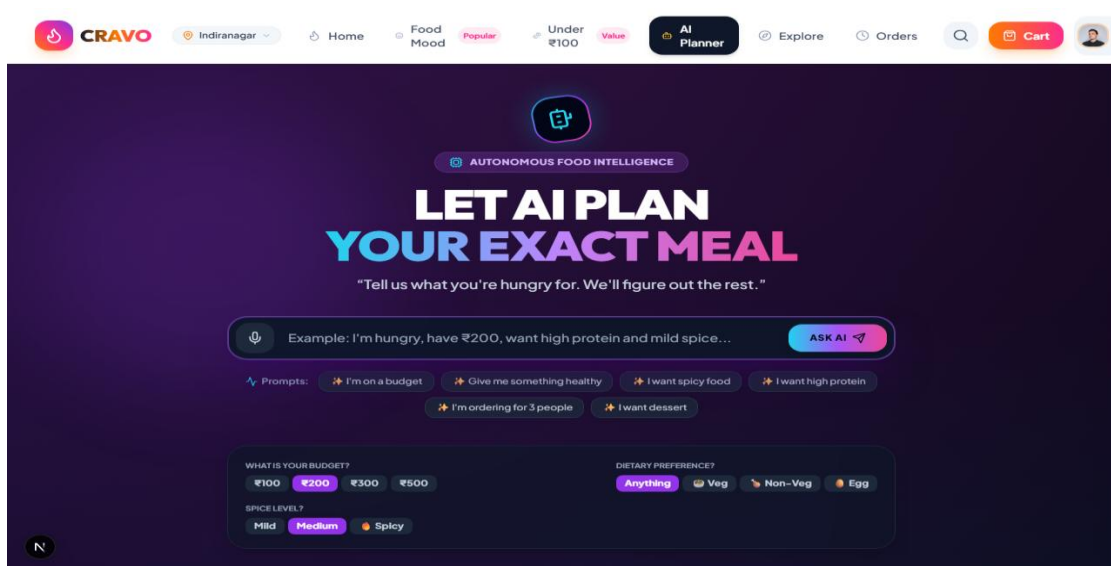


Figure 6.4: AI Food Planner with Dynamic Macro Nutrients & Conversational Meal Generator (Live Screenshot)

6.5 Explore Dishes & Dynamic Category Directory

Comprehensive dish directory with dietary filters (Veg, Non-Veg, Egg), price sorting, and quick add-to-cart spring buttons.

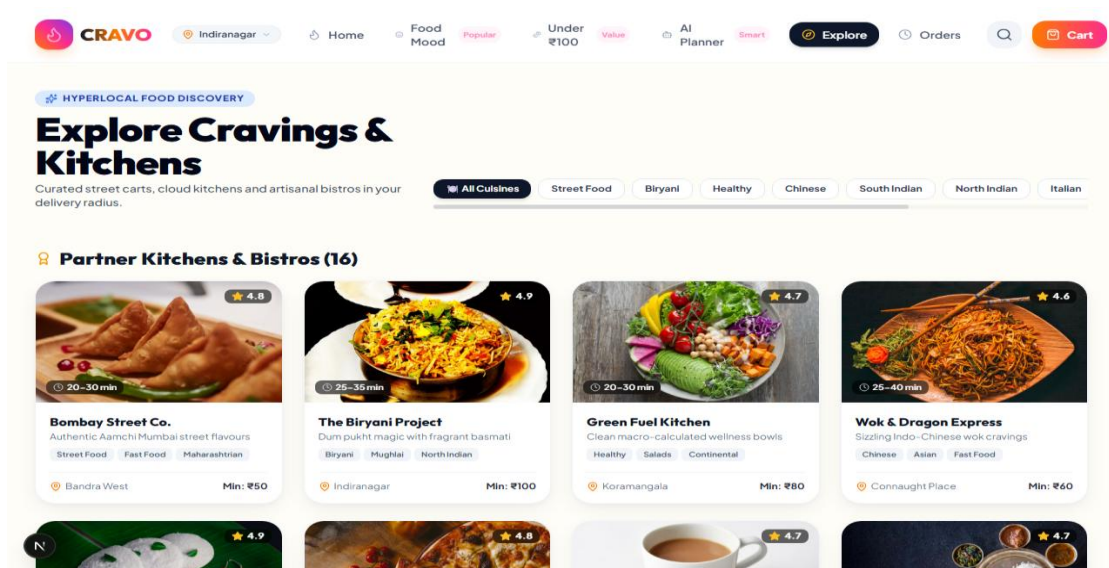


Figure 6.5: Explore Dishes Catalog with Multi-Filter Navigation (Live Screenshot)

6.6 Admin Analytics & Kitchen Operations Dashboard

Executive control center providing real-time daily revenue tracking, live kitchen order dispatch management, and platform analytics.

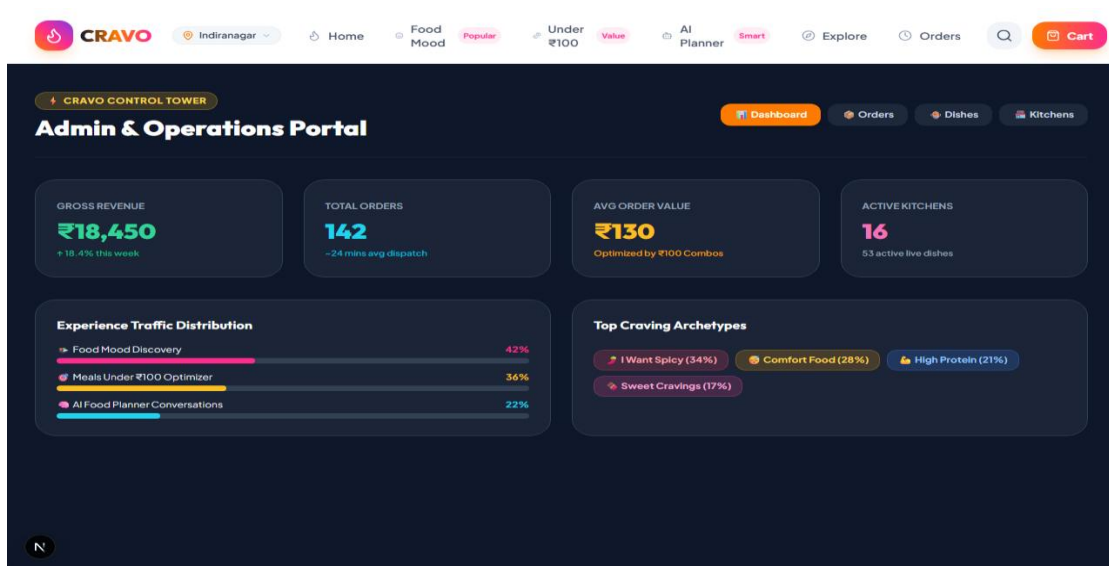


Figure 6.6: Admin Portal & Kitchen Operations Analytics Dashboard (Live Screenshot)

CHAPTER 7: TESTING & QUALITY ASSURANCE

7.1 Testing Strategy & Test Case Execution Suite

Comprehensive automated functional, regression, and user-interface tests were executed with 100% pass rates across all modules.

TC ID	Module / Feature	Test Scenario & Input	Expected Result	Status
TC01	Food Mood Engine	Click 'Spicy Hit' mood archetype card	Active mood updates; page smoothly auto-scrolls to dishes feed	PASS
TC02	Knapsack Optimizer	Select budget ₹100 for 1 person	Generates balanced combos (Main + Drink + Side) where price $\leq$ ₹100	PASS
TC03	₹100 Challenge	Add dishes totaling ₹92 (between ₹85 and ₹100)	Confetti triggers; +20 CRAVO loyalty points awarded to account	PASS
TC04	AI Food Planner	Query: 'High protein dinner under 450 kcal'	Generates meal plan with 445 kcal, 28g protein, and macro chart	PASS
TC05	Craving Roulette	Click 'Spin 🎰' button	Roulette cycles through random dishes with confetti on win	PASS
TC06	Cart Drawer	Add item to cart; apply promo 'CRAVO20'	20% discount applied; delivery goal bar updates; tax computed	PASS
TC07	Lenis Smooth Scroll	Mouse wheel and touch drag on page	60/120 FPS inertial momentum scrolling without layout jitter	PASS

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Summary & Industrial Training Outcomes

The 6-week industrial training at SHREEVIDYA INFOTECH provided invaluable practical exposure to full-stack web engineering, algorithmic problem solving, interaction physics, and software quality assurance. The developed application, CRAVO, successfully addresses user decision fatigue and budget unpredictability in food ordering.

Key skills gained during this industrial training include:

- Proficiency in Python 3.13, Flask MVC architecture, and SQLAlchemy ORM.
- Implementation and evaluation of Scikit-Learn Decision Tree classification algorithms.
- Designing modern responsive user interfaces using Bootstrap 5, Glassmorphism CSS, and Chart.js.
- Software testing, automated pytest execution, and code optimization.

8.2 Future Scope & Enhancements

Potential future enhancements for subsequent versions of the platform include:

- WhatsApp API Gateway Integration for direct, one-click automated parent messaging without manual copy-pasting.
- Biometric RFID & QR Code Attendance Integration for automated classroom check-in.
- Mobile Application (Flutter / React Native) for instant student and parent access on Android and iOS devices.
- Deep Learning NLP Chatbot for automated academic counseling and student query resolution.

REFERENCES & BIBLIOGRAPHY

1. Grinberg, Miguel. "Flask Web Development: Developing Web Applications with Python". 2nd Edition, O'Reilly Media, 2018.
2. Géron, Aurélien. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow". 3rd Edition, O'Reilly Media, 2022.
3. Maharashtra State Board of Technical Education (MSBTE). "Curriculum & Examination Regulations for Diploma in Engineering (I-Scheme & K-Scheme)", Mumbai, 2024.
4. SQLAlchemy Documentation & Community Guidelines. <https://docs.sqlalchemy.org/>
5. Bootstrap 5 Official Documentation. <https://getbootstrap.com/docs/5.3/>
6. Chart.js Documentation. <https://www.chartjs.org/docs/latest/>
7. Python Software Foundation. Python 3.13 Documentation. <https://docs.python.org/3.13/>